

HITIKA HEMNANI

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EDUCATION

University Of Southern California, Los Angeles, California

Aug 2024-May 2026

Master of Science in Computer Science

(GPA 3.6/4)

Relevant coursework: Machine Learning for Data Science, Analysis of Algorithm, Natural language processing

Vivekanand Education Society's Institute of Technology, Mumbai, India

June 2018-May 2022

Graduated with Bachelor of Engineering in Computer Science with *Distinction*

(CGPA 9.68/10)

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, and Database Management Systems

TECHNICAL SKILLS

- Programming Languages: Java and Python | Software: AutoCAD.
- Web development: HTML, CSS, JavaScript, Git, and APIs
- Operating Systems: Microsoft's Windows and Linux.

PROFESSIONAL EXPERIENCE

University of Southern California, Los Angeles, CA

Research Assistant – Department of Computer Science

Feb 2025 – Present

Mental Health Project

- Conducting research on leveraging EEG brainwave data to analyze emotional states, aiming to enhance mental health diagnostics through AI-driven models.
- Exploring deep learning architectures to classify emotional states (e.g., happy, sad, angry) based on EEG signals, with a focus on model interpretability and mental health applicability.

LTI Mindtree, Mumbai, India

June 2022 - March 2024

Technology Consultant – SAP Business Unit

Colgate-Palmolive Project

- Customized and configured Employee Central modules, including data models, workflows, and reporting to align with Colgate's HR policies and global operational standards, streamlining HR processes across multiple regions.
- Collaborated closely with Colgate's HR and IT teams, providing training and support to ensure successful user adoption and system maintenance post-rollout.

Qatalum Project (Qatar Aluminium)

- Facilitated seamless data migration by mapping fields from SAP templates to Employee Central Entity and conducting CPI testing, ensuring accurate data upload and system integration for Qatar's leading aluminium company.

BIC Corporate Project

- Analyzed and interpreted client needs, driving enhancements to the SAP Performance and Goal Management system, which significantly improved HR performance tracking and workforce management.

Trivia Software's, Thane, India

May 2021 - June 2021

Web Developer

- Constructed and crafted a robust web application using Django Framework integrating model, forms, and email-based authentication functions; deployed the project on pythonanywhere.com, culminating in the TODO app.
- Completed the entire project single-handedly and submitted the same one month before the stipulated time.

ACADEMIC PROJECTS AND PAPERS

Named Entity Recognition using BiLSTM & GloVe | PyTorch, NLP, Deep Learning

- Built a Named Entity Recognition (NER) model using a BiLSTM network with ELU activation, dropout, and GloVe embeddings. Extended the model to include CNN layers for improved feature extraction and accuracy. Tuned hyperparameters for optimized performance

Transfer Learning for Waste Image Classification | Python, Keras, TensorFlow, OpenCV

- Implemented multiple transfer learning models (ResNet50, EfficientNetB0, VGG16) to classify 9 categories of waste images. Applied data augmentation, batch normalization, L2 regularization, and early stopping for generalization. Achieved robust performance through fine-tuning and performance tracking across training and validation.

Down syndrome Detector | Python, ML, SQL, and Django - [link](#)

- Built a DCNN-based model to detect Down Syndrome in newborns through facial recognition with 79% accuracy and developed a support platform connecting families to nearby schools, hospitals, and NGOs.

EXTRACURRICULAR ACTIVITIES

Student Worker, USC Auxiliary Services

Managed event invoicing, billing audits, and vendor communications using Workday and Salesforce while supporting large-scale conference logistics-Student Worker, USC Auxiliary Services

Research Publications

Co-authored two peer-reviewed papers on AI in healthcare and smart systems, including a Down Syndrome detector and an intelligent canteen management platform

Technical Achievements

Completed JPMorgan's virtual software engineering program focused on data visualization.